

CHAOYI ZHOU

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EDUCATION

Clemson University, US <i>Ph.D in Computer Science</i> Advisor: Prof. Siyu Huang GPA: 4.00 / 4.00	<i>Jan. 2024 - Present</i>
University of Southern California (USC), US <i>M.S. in Computer Science (General)</i> Advisor: Prof. Yajie Zhao GPA: 3.60 / 4.00	<i>Jan. 2021 - Dec. 2022</i>
Nanjing University Of Posts And Telecommunications (NUPT), China <i>B.S. in Computer Science and Technology</i> GPA: 3.69 / 5.00	<i>Sep. 2016 - Jun. 2020</i>

AWARDS

Rising Researcher Award from Clemson University	<i>Apr. 2025</i>
ICLR 2025 Student Travel Award	<i>March. 2025</i>

PUBLICATIONS

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- **Chaoyi Zhou***, Xi Liu*, Feng Luo, Siyu Huang. Latent Radiance Fields with 3D-aware 2D Representations. *ICLR 2025*.
 - Xi Liu*, **Chaoyi Zhou***, Nanxuan Zhao, Siyu Huang. Bézier Splatting for Fast and Differentiable Vector Graphics Rendering. *NeurIPS 2025*.
 - Xi Liu*, **Chaoyi Zhou***, Siyu Huang. 3DGS-Enhancer: Enhancing Unbounded 3D Gaussian Splatting with View-consistent 2D Diffusion Priors. *NeurIPS 2024* **[Spotlight]**.

RESEARCH EXPERIENCE

FF3R: Feedforward Feature 3D Reconstruction from Unconstrained views (Under review)

Research Intern, Microsoft

Supervisor: Dr. Yiqi Zhong (Senior Researcher, Microsoft)

- Proposed FF3R, a fully self-supervised feed-forward framework for unified 3D geometry reconstruction and semantic reasoning from unconstrained multi-view images.
- Eliminated the need for camera poses, depth supervision, or semantic labels by relying solely on RGB and feature rendering supervision.
- Addressed global semantic inconsistency and local structural inconsistency via a Token-wise Fusion Module and a Semantic-Geometry Mutual Boosting mechanism.

Bézier Splatting for Fast and Differentiable Vector Graphics Rendering (**NeurIPS 2025**)

Graduate Research Assistant

Supervisor: Prof. Siyu Huang

- Introduced Bézier Splatting, a differentiable vector graphics representation enabling fast and high-fidelity rasterization.

- Modeled vector primitives by sampling 2D Gaussians along Bézier curves, providing accurate positional gradients at object boundaries.

Latent Radiance Fields with 3D-aware 2D Representations (ICLR 2025)

Graduate Research Assistant

Supervisor: Prof. Siyu Huang

- Introduced Latent Radiance Fields (LRF), a 3D-aware latent representation that enables geometry-aware novel view synthesis directly in the 2D latent space.
- Decoupled 3D reconstruction from the image space, enabling efficient and photorealistic rendering for unbounded scenes.

3DGS-Enhancer: Enhancing Unbounded 3D Gaussian Splatting with View-consistent 2D Diffusion Priors (NeurIPS 2024 Spotlight)

Graduate Research Assistant

Supervisor: Prof. Siyu Huang

- Leveraged video diffusion priors to enforce cross-view consistency in unbounded 3D Gaussian Splatting.
- Reformulated multi-view consistency as a temporal alignment problem in video generation.
- Achieved significant improvements in rendering quality and view consistency.

Additional Research Experience (Selected)

Student Researcher / Research Assistant

- **Outdoor Scene Reconstruction and Understanding:** Developed SfM pipelines, visualization tools, and camera SDKs for outdoor scene reconstruction.
- **Aerial-to-Ground Novel View Generation:** Built synthetic and real-world datasets using Unreal Engine and developed generative models with NeRF and diffusion techniques.
- **Monocular Depth Estimation and 3D Motion Detection:** Designed multi-scale models for depth and surface normal prediction and applied SfM for 3D localization.

TECHNICAL SKILLS

Programming:	Python, C++, CUDA
ML Frameworks:	PyTorch, JAX, TensorFlow
Vision & Graphics:	3D Gaussian Splatting, NeRF, Differentiable Rendering, OpenCV, Open3D, PCL
Generative Models:	Diffusion Models, GANs
Simulation & Data:	Unreal Engine, Unity, AirSim